ARJUNA JEE 2027

Lecture -05 | Mathematics

PERMUTATIONS AND COMBINATIONS

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Topics *to be covered*

A **Uses of nCr**

B **Gap & Tie Method**

C **Question Practice**

D



H.W. Discussion

The number of 7-digits numbers which are multiples of 11 and are formed using all the digits 1, 2, 3, 4, 5, 7 and 9 is _____.

Reattempt

Sum = 31.

$\underset{*}{a}\ \underset{-}{b}\ \underset{*}{c}\ \underset{-}{d}\ \underset{*}{e}\ \underset{-}{f}\ \underset{*}{g}$

$(a+c+e+g)-(b+d+f) = 0, \checkmark\ \pm 11, \times\ \pm 22, \times\ \pm 33 \ldots\ldots$

$21 \qquad 10$

C-I: (3, 4, 5, 9) (7, 2, 1)

II: (2, 3, 7, 9) (5, 4, 1)

III: (1, 4, 7, 9) (5, 3, 2)

$10 \qquad 21$

IV: (1, 2, 3, 4) (5, 7, 9)

$\underset{*}{_}\ \underset{-}{\downarrow}\ \underset{*}{_}\ \underset{-}{\downarrow}\ \underset{*}{_}\ \underset{-}{\downarrow}\ \underset{*}{_}$

$4! \times 3! = 144.$

$= 144$

$= 144$

$= 144$

576 fnf.

Question [JEE Mains-2023]

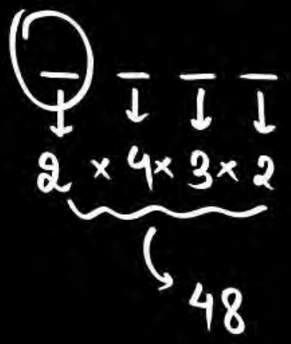
The number of integers, greater than 7000 that can be formed, using the digits 3, 5, 6, 7, 8 without repetition, is.

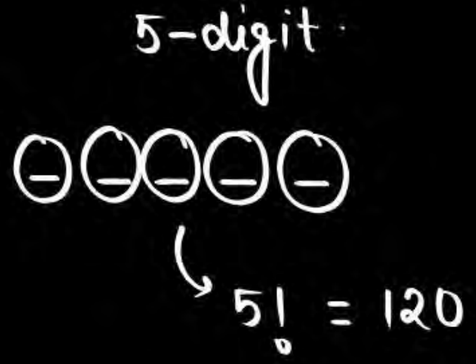
A 120

B 168

C 220

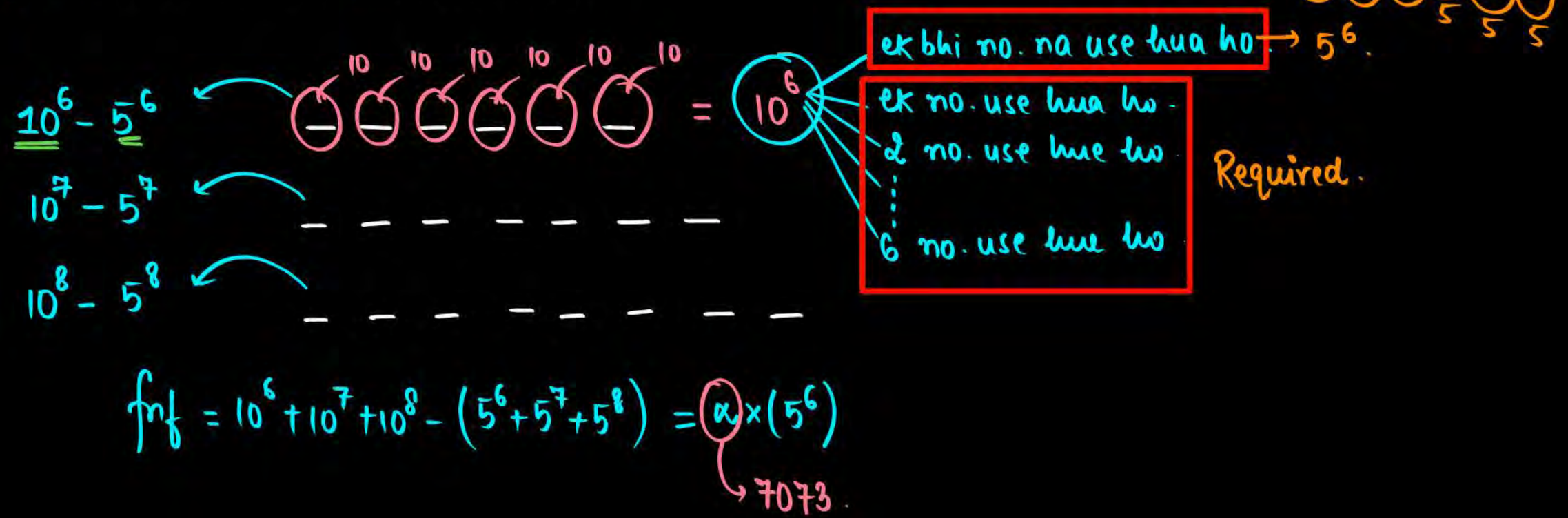
D 48





Question [JEE Mains-2022]

Let S be the set of all passwords which are six to eight characters long, where each character is either an alphabet from $\{A, B, C, D, E\}$ or a number from {1, 2, 3, 4, 5} with the repetition of characters allowed. If the number of passwords in S whose at least one character is a number from {1, 2, 3, 4, 5} is $\alpha \times 5^6$, then α is equal to



$$\text{fnf} = \underbrace{10^6 + 10^7 + 10^8} - \left(5^6 + 5^7 + 5^8\right) = (\alpha) \times (5^6)$$

$\alpha \rightarrow 7073.$

$$10^6\left(\underline{1} + 10 + 100\right) - 5^6\left(\underbrace{1 + 5 + 5^2}\right)$$

$$\underset{\uparrow}{2^6}.\underset{=}{5^6}\,(111) \quad \underset{=}{-5^6} \times 31$$

$$5^6\left(64 \times 111 - 31\right)$$

Question

If ${}^{2n}P_4 = 396 \times {}^{n}P_2$, then the value of n, is

A 4

B 5

C 6

D 7

$$\frac{(2n)!}{(2n-4)!} = 396 \times \frac{n!}{(n-2)!}$$

$$\frac{2n\cdot(2n-1)\cdot(2n-2)\cdot(2n-3)\cdot(2n-4)!}{(2n-4)!} = 396 \times \frac{n(n-1)\cdot(n-2)!}{(n-2)!}$$

$2(n-1)$

$(2n-1)(2n-3) = 11 \times 9$

$2n-3 = 9$

$n = 6$

Question [JEE Mains-2019]

A committee of 11 members is to be formed from 8 males and 5 females. If m is the number of ways the committee is formed with at least 6 males and n is the number of ways the committee is formed with at least 3 females, then

(A) $m = n = 68$

(B) $m + n = 68$

(C) $m = n = 78$

(D) $n = m - 8$

8M, 5F

$m = {}^8C_6\,{}^5C_5 + {}^8C_7\,{}^5C_4 + {}^8C_8\,{}^5C_3 = 78$

(6M 5F, 7M 4F, 8M 3F)

$n =$ (8M 3F, 7M 4F, 6M 5F)

Example

A dice is rolled 3 times, top-most face is considered as outcome then find the number of outcome such that :

(i) 5 appears atleast once.

(ii) All 3 outcomes are different.

(iii) Exactly 2 outcomes are same.

(i) $\underline{6}\ \underline{6}\ \underline{6}$

Total $= 6^3$.

atleast once = Total − (zero times)

$= 6^3 - (5^3)$

$= 91$.

ex:

325
164
235
⋮

$6 \times 5 \times 4 = 120$.

(iii) ex:-

225
141
233
616
441
⋮

(iii)

Total $= 6^3$

- all different $= 120$
- exactly 2 out. are same $= 216 - (126) = 90$
- all same $= 6$.

OR

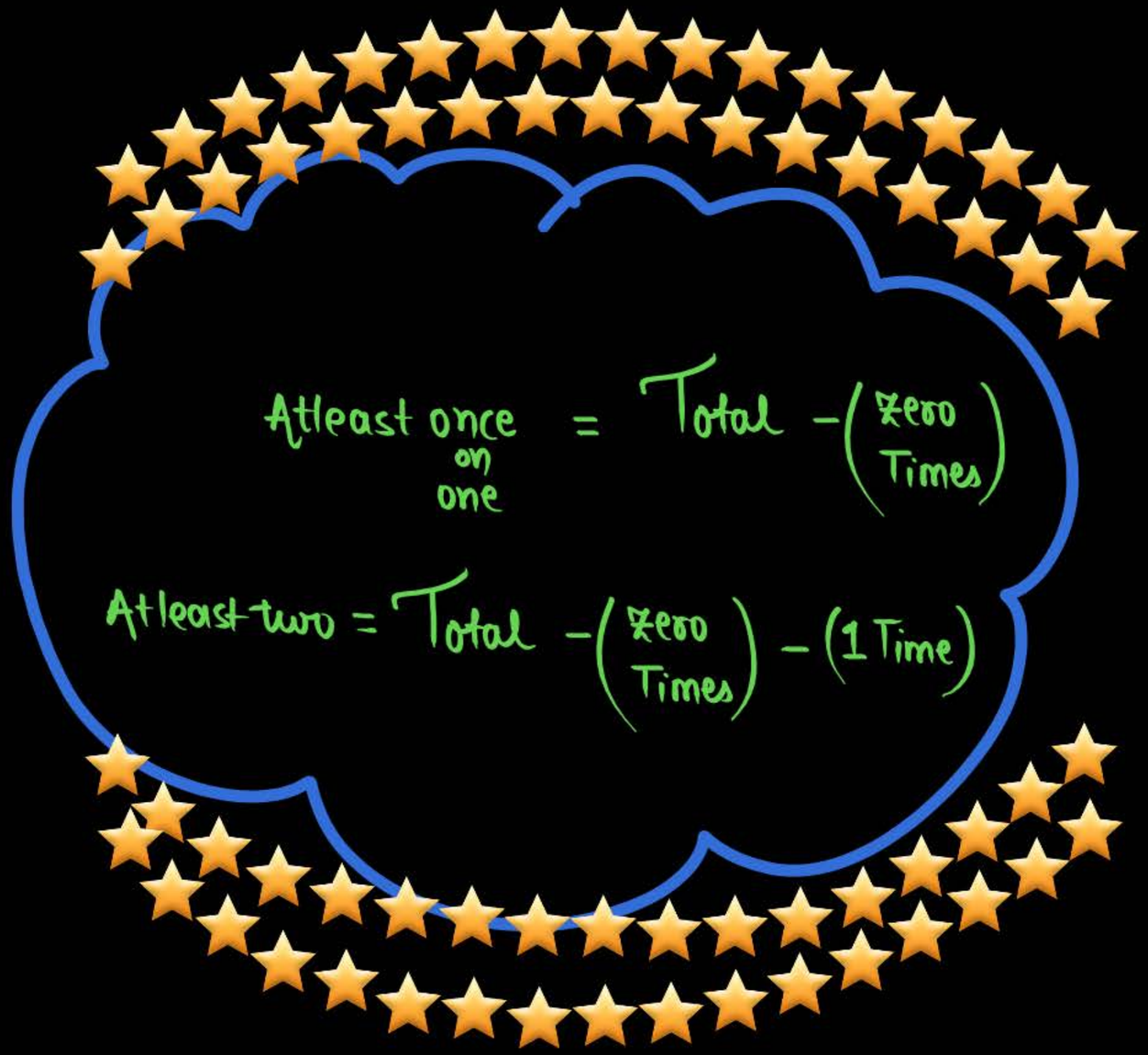
$\{1, 2, 3, 4, 5, 6\}$



select any 2 $\times$ no. of outcomes.

${}^6C_2 \times 6$

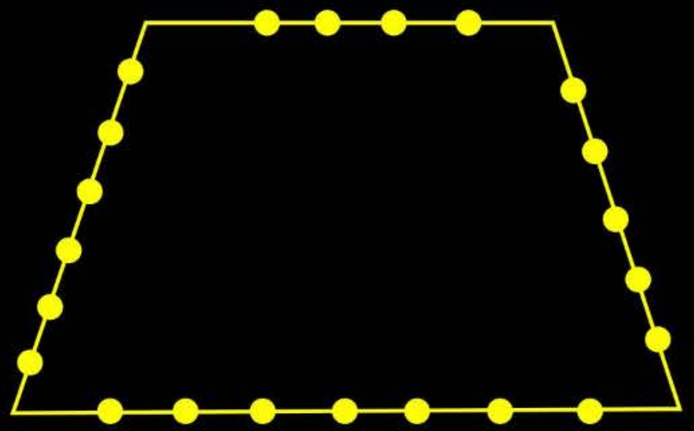
$15 \times 6 = 90$


Atleast once on one = Total - (Zero Times)
Atleast two = Total - (Zero Times) - (1 Time)

Question

(i) Number of Straight lines.

(ii) Number of Δ's.



$$^{22}C_2 - \left(^5C_2 + {}^4C_2 + {}^7C_2 + {}^6C_2\right) + 4$$

$$231 - (10 + 6 + 21 + 15) = 183.$$

$$^{22}C_3 - \left(^5C_3 + {}^4C_3 + {}^6C_3 + {}^7C_3\right) = 1471.$$

Total points $= 7+6+5+4$

$= 22.$



IMPORTANT RESULTS

RAR

In '*n*' sided polygon the triangles are formed by joining the vertices of polygon then:

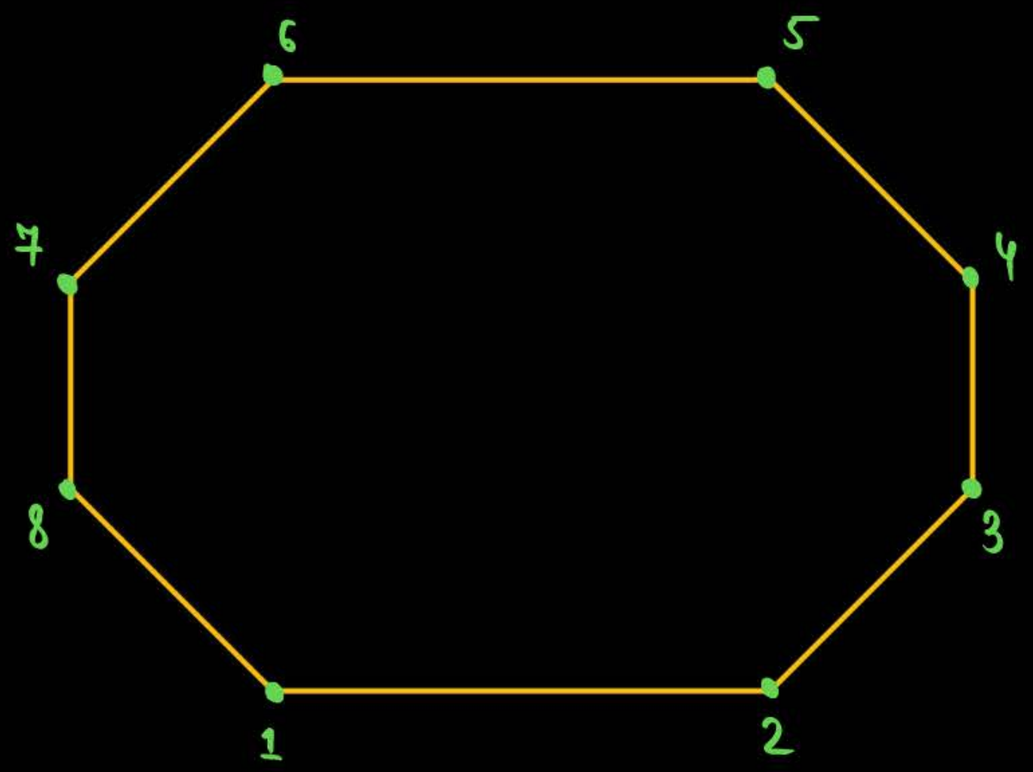
(a) Number of triangles having one side common with the sides of polygon = $n(n-4)$

(b) Number of triangles having two side common with the sides of polygon = n.

(c) Number of triangles having no side common with polygon = $^{n}C_{3} - (n(n-4) + n)$

(d) Number of diagonals in 'n' – sided polygon = $\boxed{^{n}C_{2} - (n)}$

CONSIDER OCTAGON $(n=8)$



(i) no. of diagonals $= {}^{8}C_{2} - (\overset{\text{sides}}{8})$

(ii) no. of Δ's having 1 side common with Octagon $= 8 \times 4$
$= 32.$

(ii) " " " 2 " " " " " $= 8$

(iii) " " " no " " " " " $= {}^{8}C_{3} - (32+8)$

Question [JEE (Adv.)-2014/JEE Mains-2020]

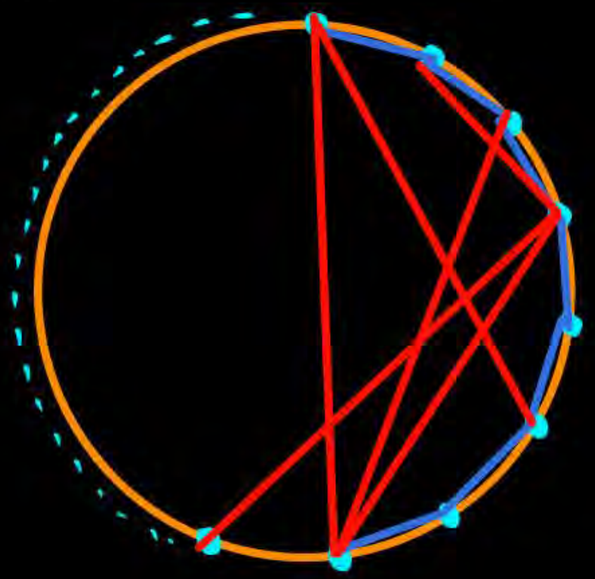
Let $n > 2$ be an integer. Suppose that there are in Metro stations in a city located along a circular path. Each pair of stations is connected by a straight track only. Further, each pair of nearest stations is connected by blue line, whereas all remaining pairs of stations are connected by red line. If the number of red lines is 99 times the number of blue lines, then the value of n is

A 199

B 201

C 101

D 200



blue lines → sides

Red line → diagonal.

diagonal = 99 sides.

$${}^{n}C_2 - n = 99n$$

$${}^{n}C_2 = 100n$$

$$\frac{n!}{2!(n-2)!} = 100n$$

$$\frac{n(n-1)(n-2)!}{2 \times (n-2)!} = 100n$$

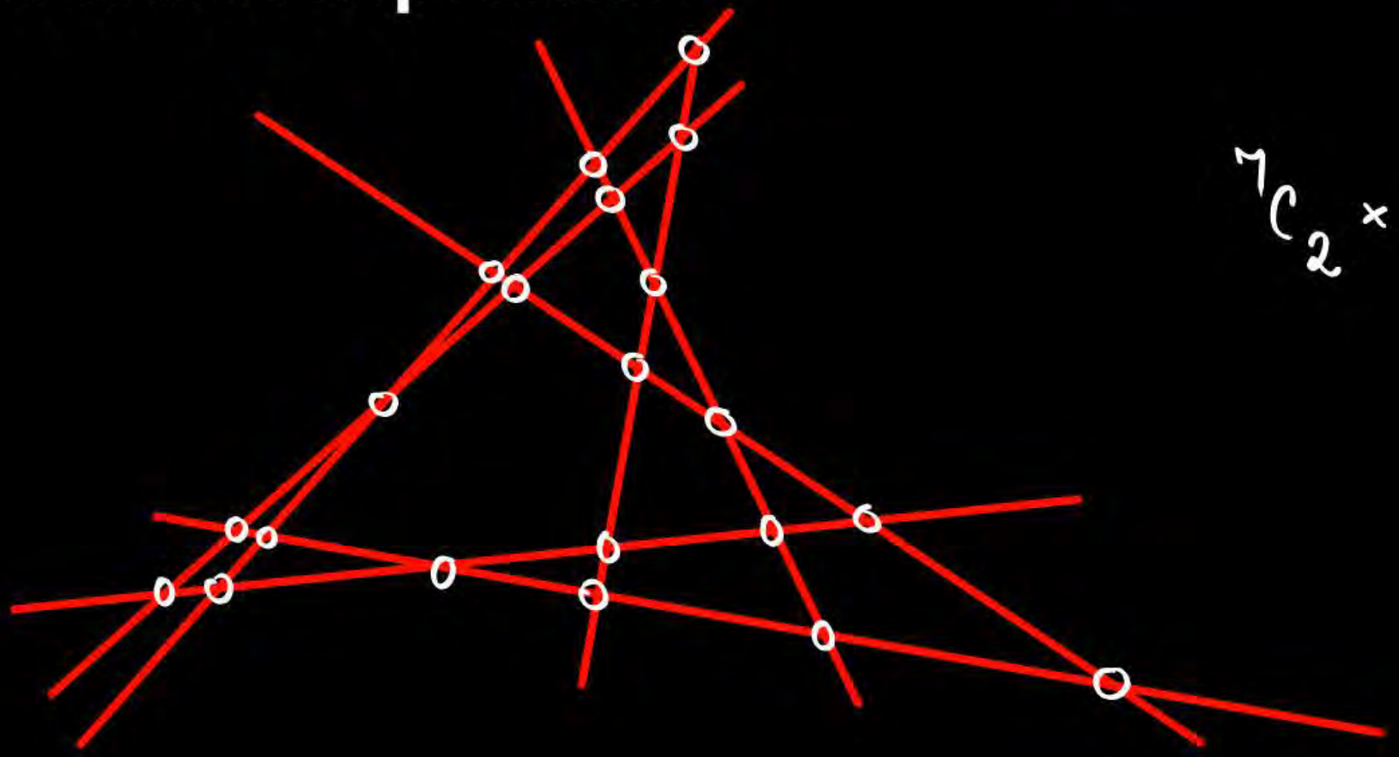
$$n-1 = 200$$

$$n = 201.$$

COUNTING POINT OF INTERSECTION

Example

There are 7 straight lines in a plane, then find maximum number of point of intersection possible.



$^{7}C_{2} \times 1 = 21.$

COUNTING POINT OF INTERSECTION

Example

There are 7 straight lines in a plane out of which 4 are concurrent, then find maximum number of point of intersection possible.

passing thru same point.

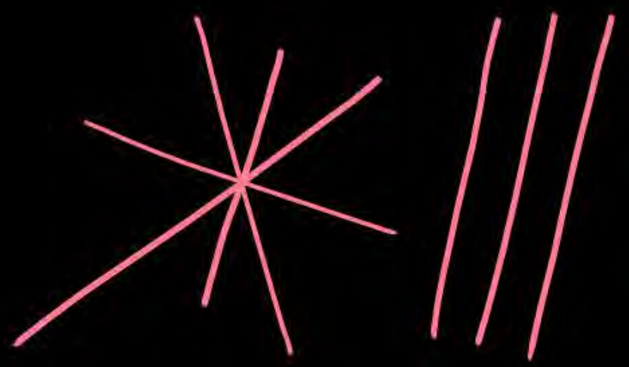
$${}^{7}C_{2} - {}^{4}C_{2} + 1$$

$$21 - 6 + 1 = 16$$

COUNTING POINT OF INTERSECTION

Example

There are 11 straight lines in a plane out of which 4 are concurrent & 3 are parallel, then find maximum number of point of intersection possible.



$$\text{max} = {}^{11}C_2 - {}^{4}C_2 - {}^{3}C_2 + 1 .$$

COUNTING POINT OF INTERSECTION

Example

There are 4 straight lines, 3 circles and 2 parabola's are present, then find maximum number of point of intersection possible.

$$\text{Total p.o.i.} = \underbrace{\left({}^{4}C_{2} \times 1\right)}_{\text{L-L}} + \underbrace{{}^{4}C_{1} \times {}^{3}C_{1} \times 2}_{\text{L-C}} + \underbrace{{}^{4}C_{1} \times {}^{2}C_{1} \times 2}_{\text{L-P}} +$$

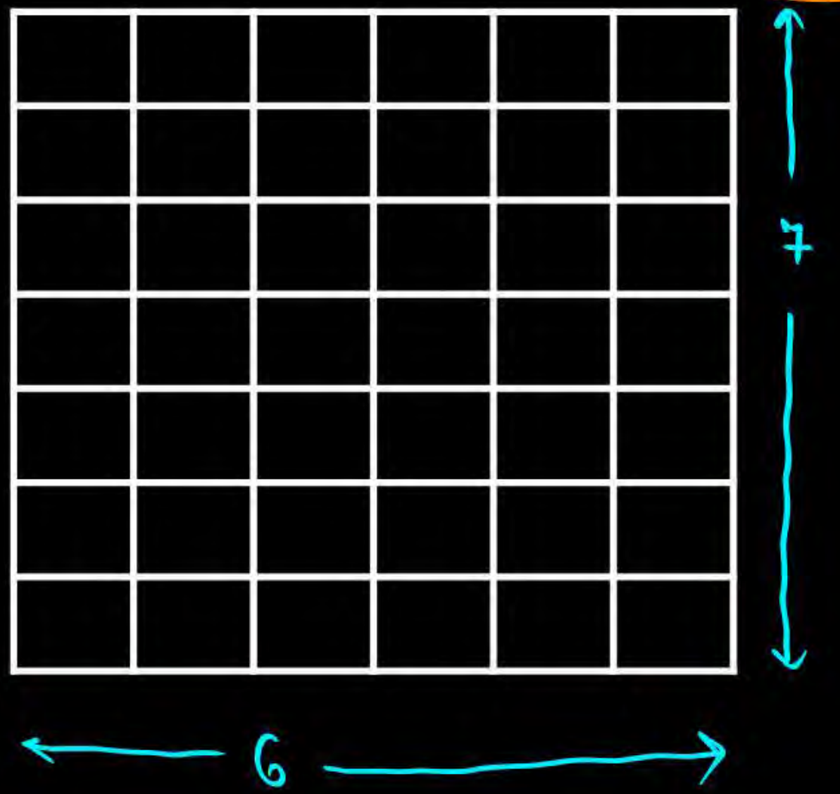
$$\underbrace{{}^{3}C_{2} \times 2}_{\text{C-C}} + \underbrace{{}^{3}C_{1} \times {}^{2}C_{1} \times 4}_{\text{C-P}} + \underbrace{{}^{2}C_{2} \times 4}_{\text{P-P}}$$

COUNTING RECTANGLES & SQUARES

Every Square is a Rectangle.

Example

Count total possible Squares and Rectangles in given arrangement.



dimension wise.

Squares of $(1\times1) = 7\times6 = 42$

" " $(2\times2) = 6\times5 = 30$

$(3\times3) = 5\times4 = 20$

$(4\times4) = 4\times3 = 12$

$(5\times5) = 3\times2 = 6$

$(6\times6) = 2\times1 = 2$

add ✓

no. of vertical lines = 7

no. of Horiz. lines = 8

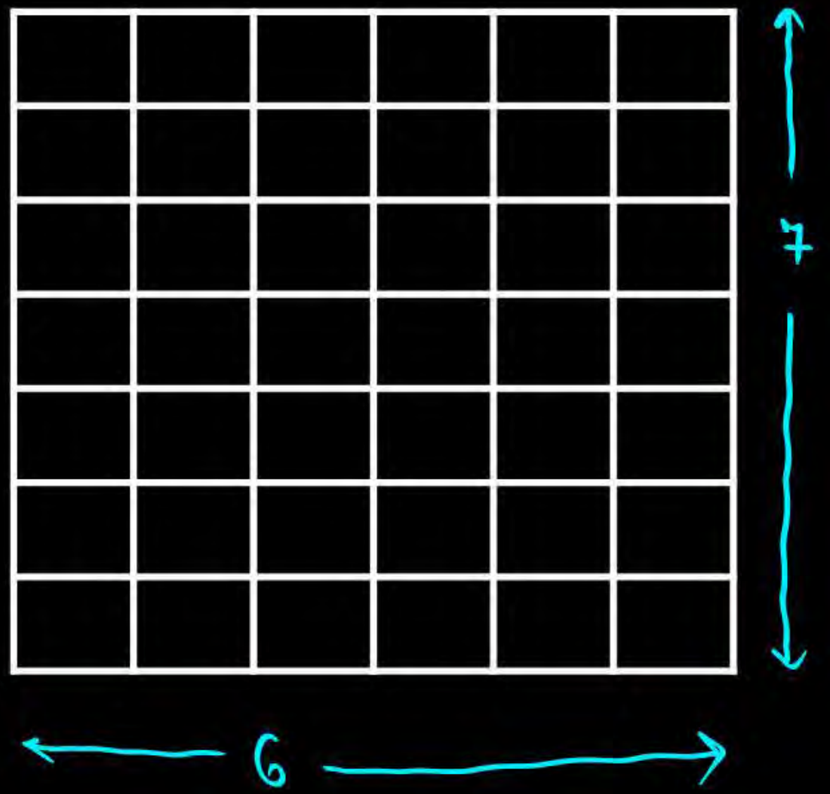
no. of Rectangles $= {}^{7}C_2 \times {}^{8}C_2$

COUNTING RECTANGLES & SQUARES

Every Square is a Rectangle.

Example

Count total possible Squares and Rectangles in given arrangement.



no. of Rectangle with (odd - dimensions): ${}^{12}C_1 \times {}^{16}C_1 = (12 \times 16)$

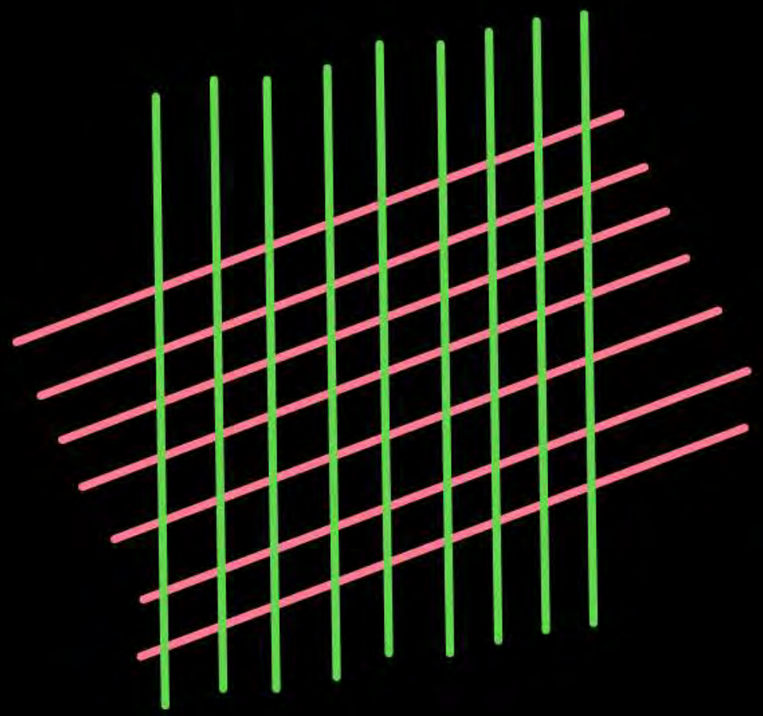
both l & b are odd.

Horizontal → 1unit → 6opts.
3unit → 4opts.
5unit → 2opts.

Total = (12)

Vertical → 1unit → 7opts.
3unit → 5opts.
5unit → 3opts.
7unit → 1opts.

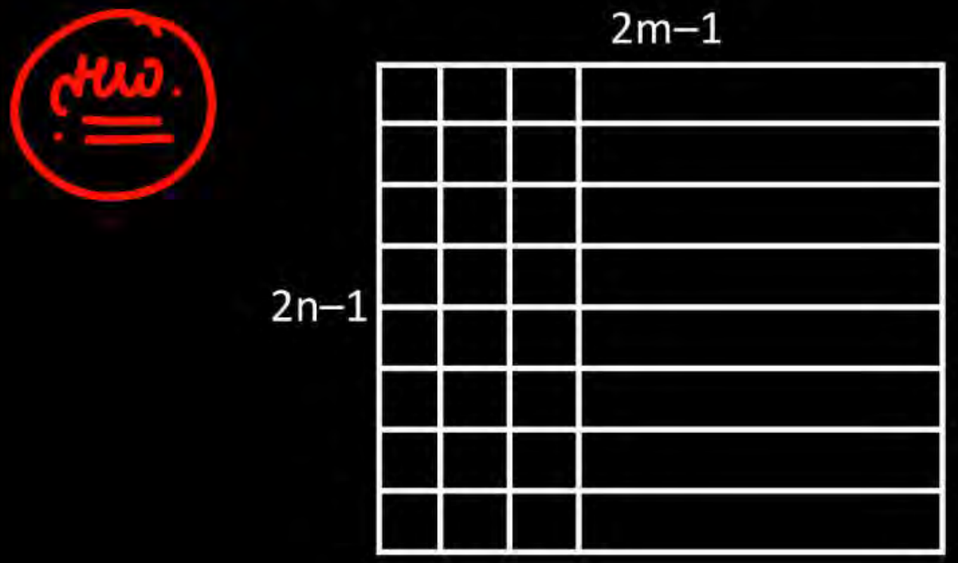
Total = 16



no. of llgms $= {}^{9}C_2 \times {}^{7}C_2$.

Question [IIT-JEE-2005]

There is a rectangular sheet of dimension $(2m-1) \times (2n-1)$, (where $m > 0$, $n > 0$). It has been divided into square of unit area by drawing line perpendicular to the sides. Find the number of rectangles having sides of odd unit length.



A $(m+n+1)^2$

B $mn(m+1)(n+1)$

C 4^{m+n-2}

D m^2n^2

Challenger Question

Total 5 different pair of shoes are there, you need to select only 4 shoes such that

(i) No restriction

(ii) Exactly 1 pair of should be formed

(iii) Exactly 2 pair of should be formed

(iv) No pair should form

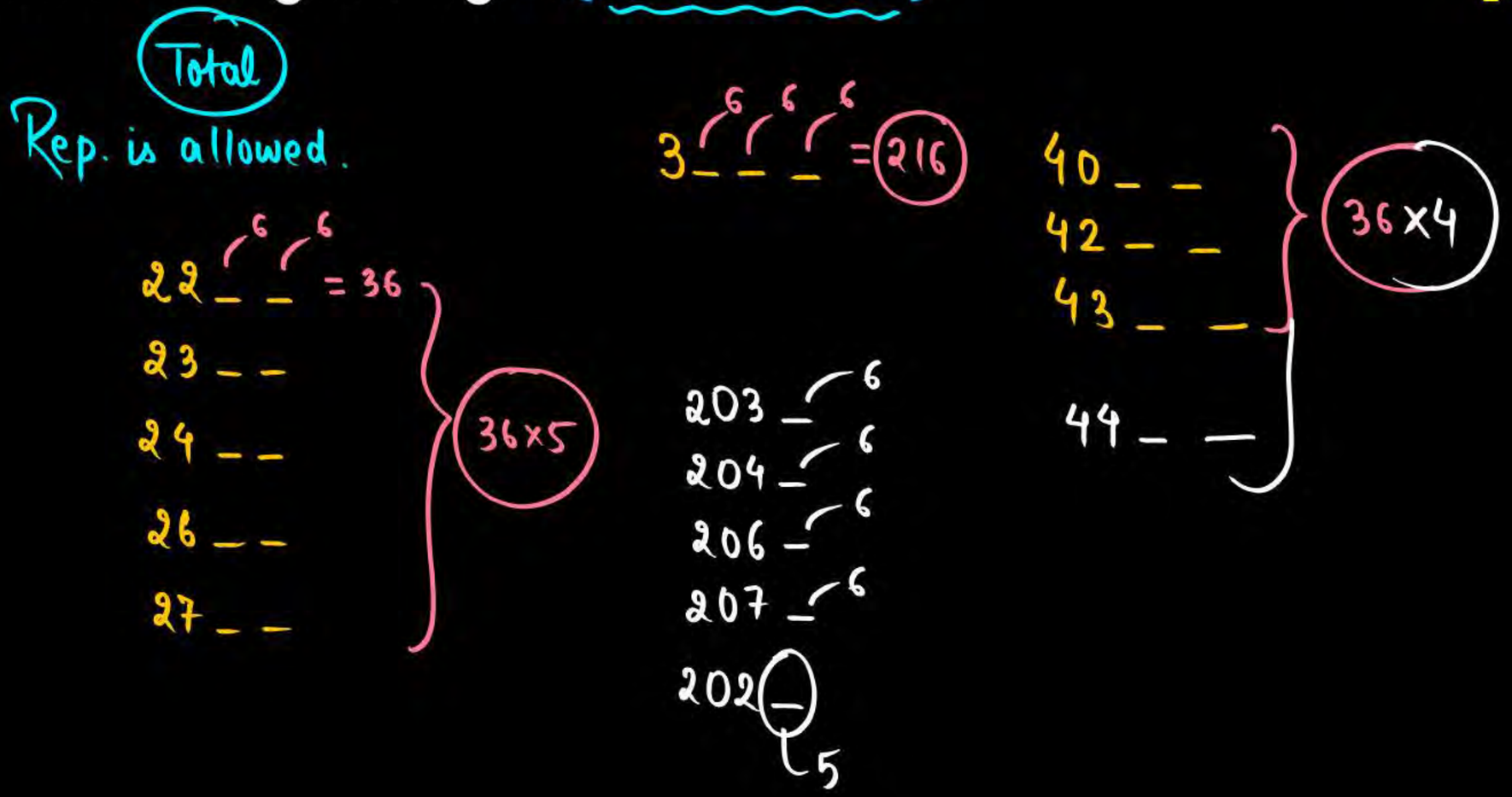


DO IT BY YOURSELF (DIBY)

DIBY-11

Find the number of 4 digit integers in the closed interval [2022, 4482] which can be formed using the digits {0, 2, 3, 4, 6, 7}. **[JEE (Adv.)-2022]**

[Ans. : 569]

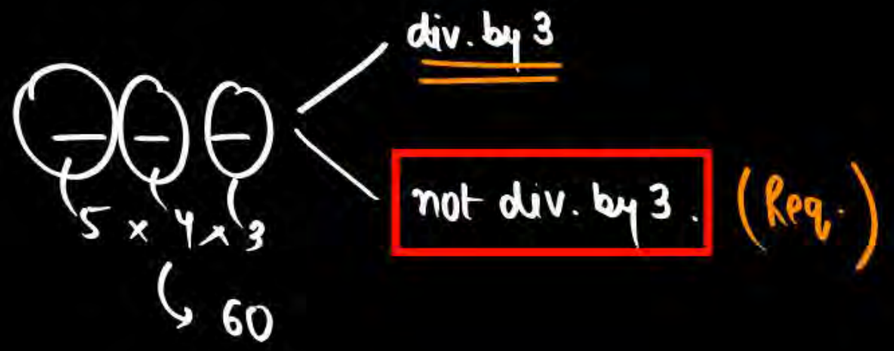


DIBY-12

The number of 3-digit numbers, formed using the digits 2, 3, 4, 5 and 7, when the repetition of digits is not allowed, and which are not divisible by 3, is equal

[JEE Mains-2024]

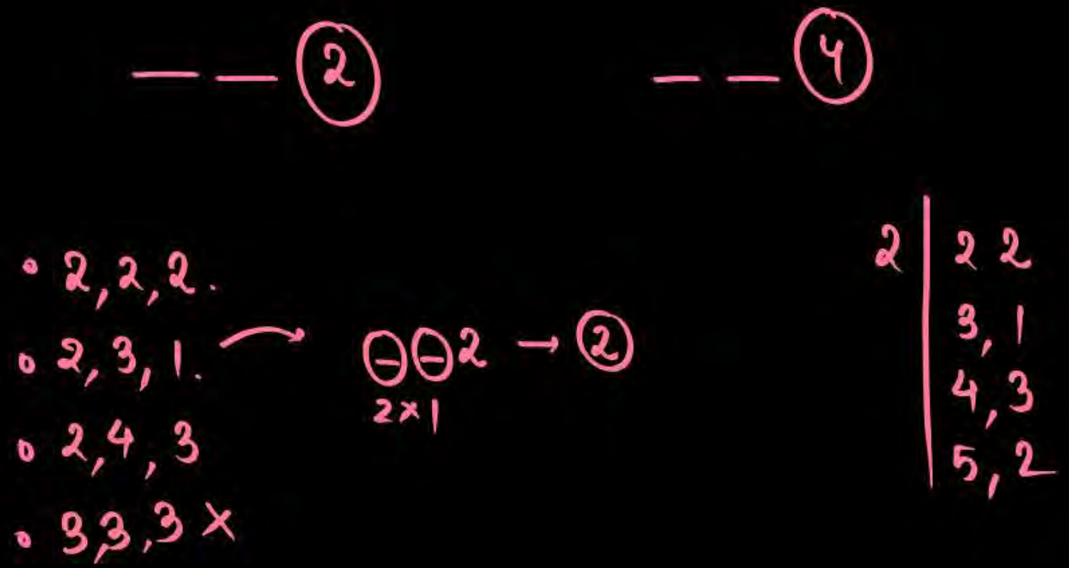
[Ans. : 36]



DIBY-13

Total numbers of 3-digit numbers that are divisible by 6 and can be formed by using the digits 1, 2, 3, 4, 5 with repetition, is _____. **[JEE Mains-2023]**

[Ans. : 16]



DIBY-14

The number of words that can be formed out of the letters of the word ARTICLE so that the vowels occupy even places is

(A) 36 (B) 574 (C) 144 (D) 754

[Ans. : (C)]

DIBY-15

Eleven books consisting of 5 Mathematics, 4 Physics and 2 Chemistry are placed on a shelf. The number of possible ways of arranging them on the assumption that the books of the same subject are all together is

(A) 4!2! (B) 11! (C) 5!4!3!2! (D) None of these

[Ans. : (C)]

DIBY-16

The number of six letter words (with or without meaning), formed using all the letters of the word 'VOWELS', so that all the consonants never come together, is

(A) 576 (B) 140 (C) 356 (D) 209

[Ans. : (A)]



Homework

Re-attempt all Class Examples

DPP & DIBY

It is never about end result

It's always about Journey

Thank You

#futureIITians